The Operation of ZeTa

Wolfgang Grieskamp

July 26, 2000 – for version 1.5 of ZeTa

Abstract

The ZeTa tool environment provides an open framework for the integration of tools and notations. Regarding its custom functionality and semantic support, it bases on the Z[1] notation. This document gives a brief overview on the operation of the ZeTa environment, addressing users working with the environment.

Contents

1 Tutorial Introduction **1**

1.1 Starting a session 2

1.2 Loading sources 2

1.3 Performing Operations 4

1.4 Browsing Locators 6

1.5 Cut and Paste 7

2 XEmacs Support for ZeTa **7**

3 Batch Operation **9**

1 Tutorial Introduction

The ZeTa system can be run in one of three operation modes:

1. from within the XEmacs editor[2];
2. with a graphical user interface (based on Java's Swing library)[3];
3. in a batch-oriented mode, where ZeTa reads its commands from standard input and echos the results to standard output

In fact, the first two modes base on the third one, which represents the "core" functionality of ZeTa. However, in this tutorial introduction, we will explain ZeTa's features along its graphical user interface.

> *Remark*: We assume that ZeTa has been successfully installed at your site. See *The Installation of* ZeTa[4] for notes on installation.

[1] z.html
[2] GNU Emacs may work as well, though this has not been tested for a while.
[3] This GUI is still experimental
[4] install.html

1.1 Starting a session

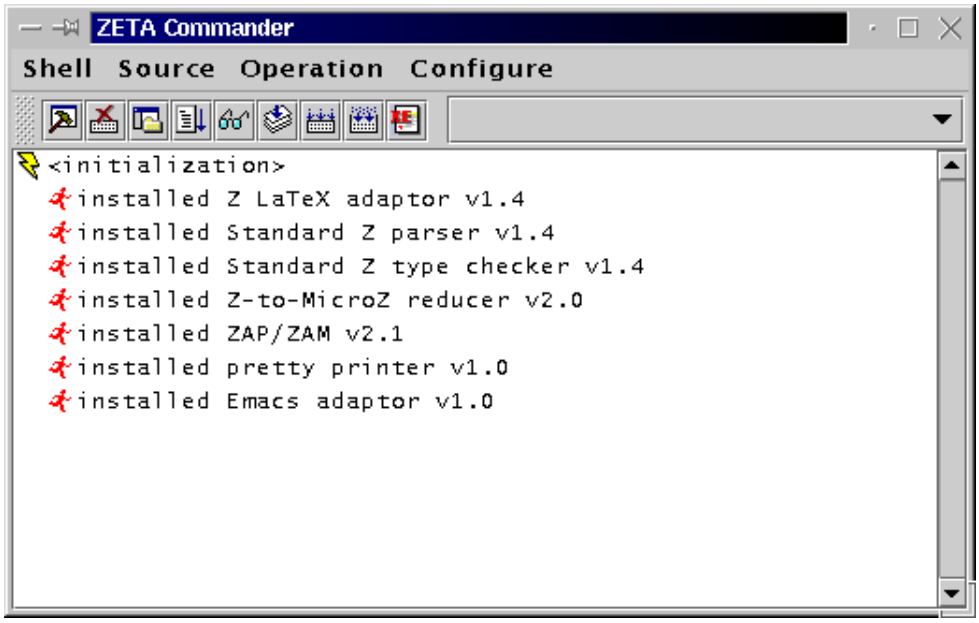
Make a working copy of the `$ZETABASE/examples` directory (where `$ZETABASE` is the installation base of ZETA). We call this copy `examples` in the sequel.

Change to the directory `examples/bbook`, which contains the well-known example of the birthday-book specification, and type `zeta&` from the shell.

> *Remark*: If the command `zeta` is not found, you may need to put `$ZETABASE/bin` in the `PATH`.

> *Remark*: Under Windows, you may run the program `runzeta.exe`, which starts with a browser for the working directory.

You should get the following window:

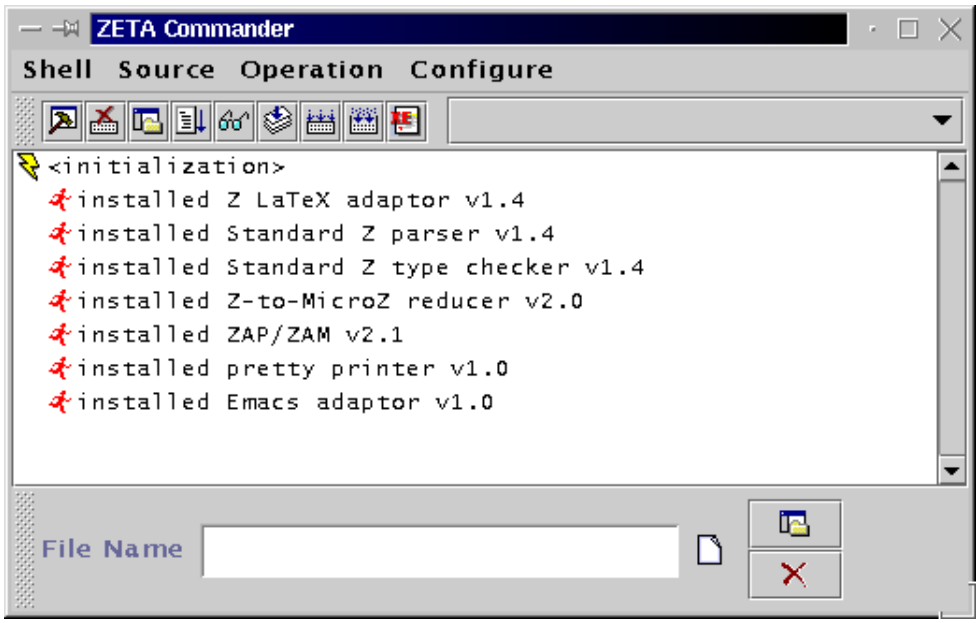


This window is the ZETA *commander*. The commander's menu allows you to access the operations of ZETA, and to launch further windows, as will be explained below. The window displays the so-called *session log*. The session log consists of *transactions* which are indicated by the little lightening icon to the left. The messages belonging to a transaction are displayed intended to a description of the transaction. In the window above, we see the transaction of initialization of ZETA, with a set of *progress messages* (these are indicated by the "running man"), telling which *plugins* have been installed for your ZETA session. The actual number of plugins depends on your ZETA setup.

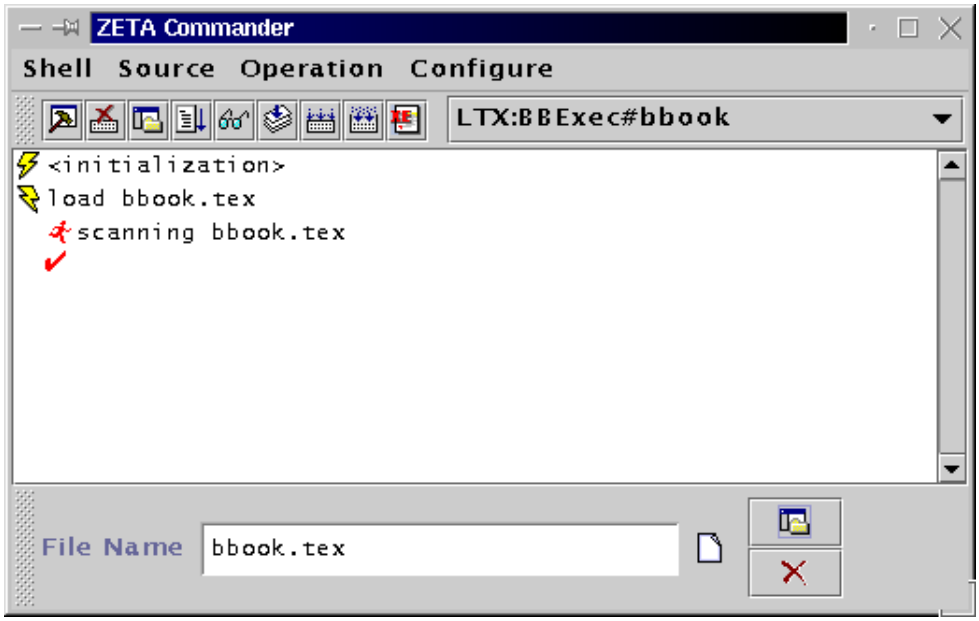
You can click on the lightening icon of a transaction to fold resp. unfold the messages belonging to this transaction.

1.2 Loading sources

Before you can work with your documents, they have to be *loaded* into the ZETA session. Documents to be initially loaded can be specified from the command line (`zeta doc1.tex doc2.tex ...`), or laterly loaded via via the `Source` menu of the commander or the tool bar. Activate `Source` ⟶ `Load LaTeX File`. You get the following modified window:



Type in `bbook.tex` into the botton panel, or select the file via the file browser. Then hit the button with the file-folder icon (note that this is the same icon which represents this function in the tool bar). The effect of loading will be displayed in the session log:



You see a progress message indicating that the LaTeX source has been scanned, and a check mark which tells you that this transaction has been successfully finished. The messages of the previous transaction has been automatically folded. They are still there, you can see them by clicking on the lightening icon.

Loading. What does the load operation conceptually means? From the time on a source has been loaded into a ZETA session, the session "takes care" of this source. It monitors modifications you make on the source (e.g. by some external editor), and whether results of operations you

have applied to the specifications contained in the source are up-to-date (e.g. type-checking, compilation, etc.). This is quite comparable to traditional Unix' `make`.

Note that you only need to load the main document of a compound document. Subdocuments included from the main document will be automatically tracked by ZETA.

Batch operation. It would be cumbersome if each time you start a ZETA session, sources need to be loaded by hand again. You can put the according load commands in a file called `zetarc` in the working directory of a project. In the `examples/bbook` directory, you find a file `zetarc.bbook` with the following content:

```
# This file contains commands executed by ZETA on startup.
# ZETA executes (if found) the following configuration files
#    $ZETABASE/lib/zetarc
#    $HOME/.zetarc
#    ./zetarc

# load the LaTeX sources of the birthday-book example:
load bbook.tex
```

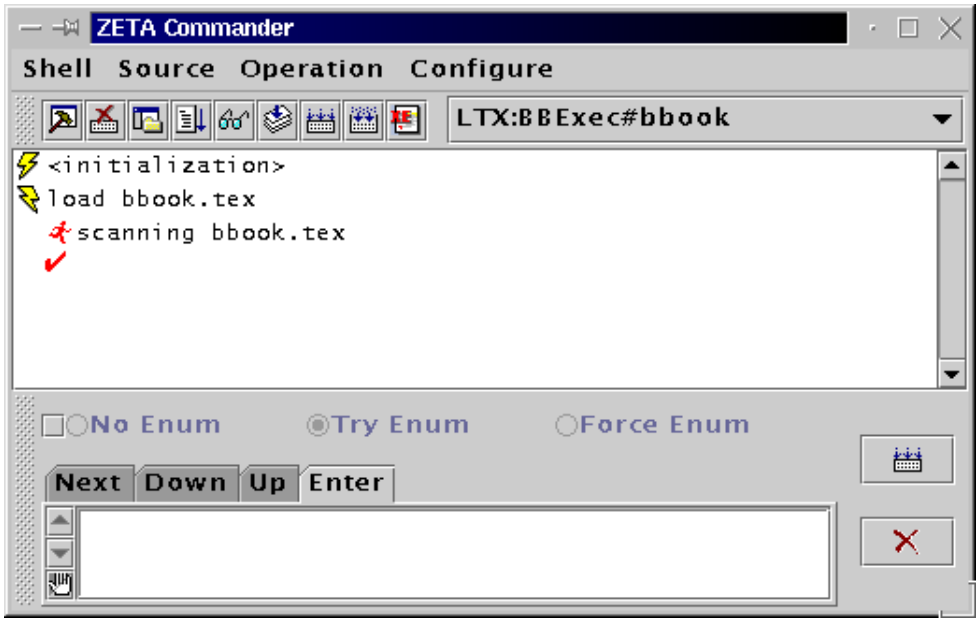
This file can be renamed to `zetarc`, so the next time a ZETA session is started from this directory, the source `bbook.tex` will be automatically loaded.

You may note that the string `load bbook.tex` is exactly the same as displayed in the commanders session log to describe the transaction of loading. In fact, any operation available from the GUI eventually leads to a command which could be as well executed in batch mode.

1.3 Performing Operations

The operations of the ZETA system can be reached via the menu `Operation` or the tool bar. The provided operations depend on which plugins have been installed for ZETA. We assume that you have at least the ESZ type-checker plugin and the ZAP compiler plugin.

Activate the `Operation` ⟶ `ExecuteExpression` menu item. You get the following window:

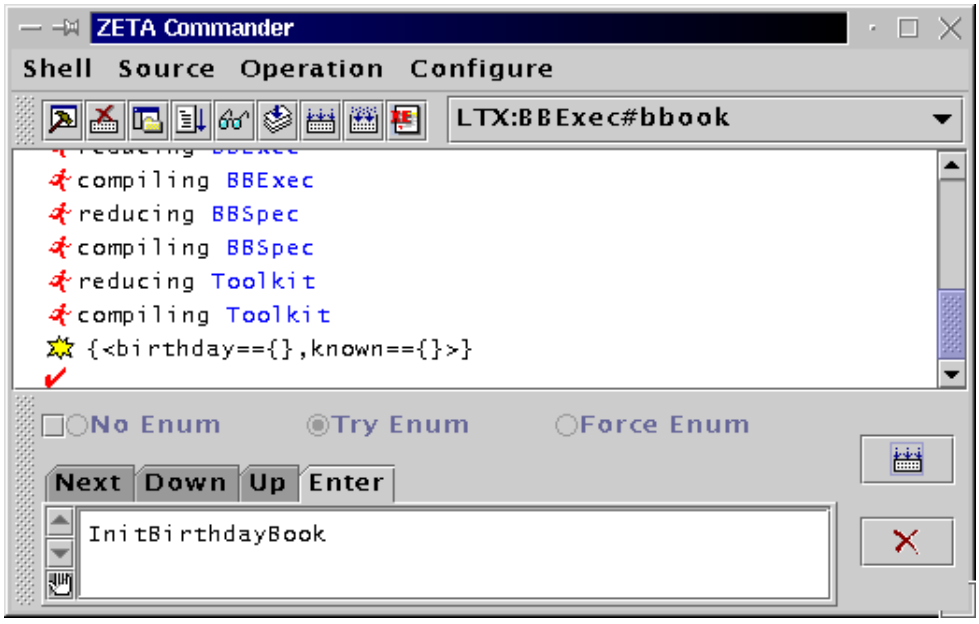


This window allows you to execute Z expressions. You can choose between evaluation modes and step through solutions which are described in the *ZAP manual*[5].

In order to perform an operation, you need to specify the Z section it targets. This is done via the combobox at the left of the tool bar. When loading a source, all available sections are scanned from the source, and are made available for operations. Clicking on the combobox shows that the source *bbook.tex* has provided several sections (see *The Z of* ZETA[6] for more information about Z sections).

> *Remark*: The sections in this example are the followings:
>
> - `LTX:BBSpec#bbook.tex`: a section containing the basic specification of the birthday book;
> - `LTX:BBExec#bbook.tex`: a section containing a refinement of the specification of the birthday book for the purpose of executability;
> - `LTX:BBTest#bbook.tex`: a section containing a refinement of the executable specification for the purpose of test-case evaluation;
>
> For more details of the usage of these sections in the birthday book example, see the birthday-book document.
>
> *Remark*: The name of a section (or, more generally, *units*) in ZETA, is given by a so-called MSL (model source locator). An MSL consists of (1) an identifier denoting the tool which has imported the unit into ZETA (here, the LaTeX-tool) (2) the name of the unit (here, e.g. `BBExec`) (3) tool specific data (here, for LaTeX, the name of the file where the section originates from).

To carry on with the example, let use choose the section `LTX:BBExec#bbook.tex`. Next type in the text field an expression, e.g. *InitBirthdayBook*, and hit the apply button. In the session log, you see some progress messages; the tail of this output will look as follows:



In order to execute the given Z expression, ZETA has operated a compound tool-chain, which you completely see if you scroll or resize the commander window: the Z toolkit this specification bases on has been loaded, several sections have been parsed and type-checked, and the section in which

[5] zap.html
[6] z.html

context the given expression is to be executed has been reduced and compiled. Finally, the result of the evaluation has been printed (`{<birthday == {}, known == {}>}`).

Lazy tool chains. Tool chains are executed "lazily" by ZeTa. If you now hit again the `Apply` button of the `Execute` panel, the result will be immediatly prompted. But if the original source `bbook.tex` is modified (e.g. by saying `touch bbook.tex` from a Unix shell), the tool-chain is recomputed.

Editing text fields. The text field of the `Execute` panel, as most other text fields in ZeTa, provides a history. Stepping through the history can be achieved by clicking on the up-arrow and down-arrow in the left upper corner of the text field.

It is possible to perform cut, copy and paste on the text field, for exchange with other applications. Note, however, that under X the Java system addresses the *system clipboard*, not the current selection. (The current selection is what is highlighted if you drag the mouse over some text displayed in an X application.)

From within a text field, you can use the keys `C-c` and `C-v` to copy into resp. paste from the system clipboard. Under XEmacs you can reach copy and paste for the system clipboard via the tool bar.

In order to execute the text of a panel from an external editor, click on the "hand" icon at the left bottom. This passes control on this text field to an externally attached editor, usually XEmacs.

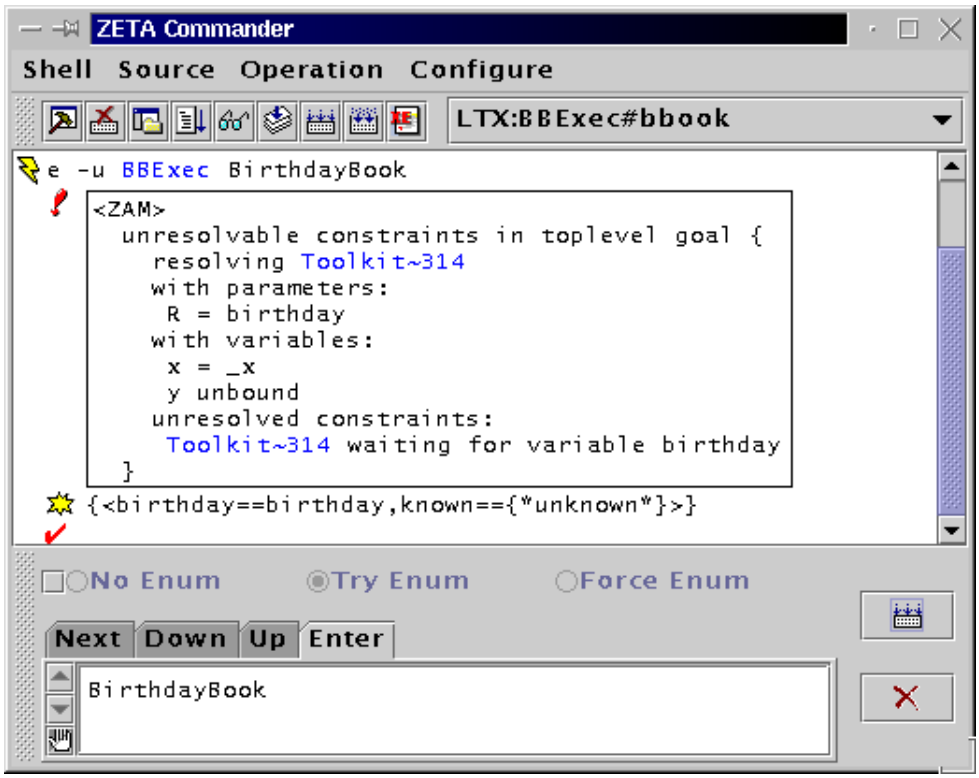
1.4 Browsing Locators

Diagnostics and other messages written in the session log may contain so-called *locators*, which point to sources, and which can be browsed by clicking on them.

> *Remark*: To let locator browsing work, you need to have XEmacs 20 or later installed. See *The Installation of* ZeTa[7] for information about installing *all* ZeTa support for XEmacs in your `.emacs`.

For trying out locators, enter e.g. the expression `BirthdayBook` into the execution panel, then hit the apply button. You get the following log:

[7] install.html



This log contains an error diagnostic which describes the failure of the execution, caused by the attempt to enumerate the bindings of the schema *BirthdayBook*. You can click on the blue locators to visit the sources referred to from the diagnostic. If XEmacs is already running, it will be contacted to visit the file and highlight in this file the region designated by the locator. If not, it will launched.

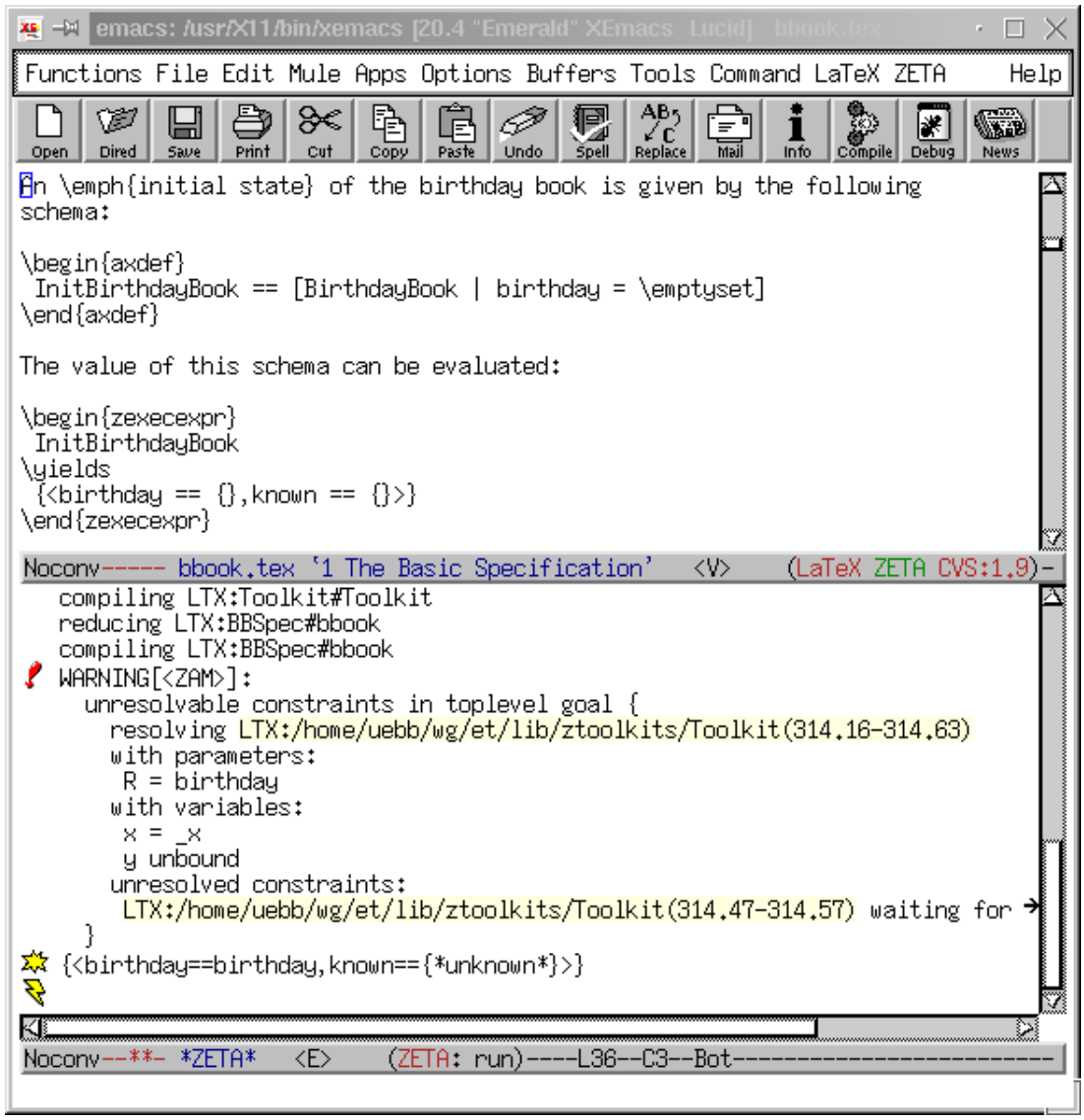
1.5 Cut and Paste

The Java based GUI of ZETA does not implement special facilities for cut and paste, but takes the defaults of the Java implementation. You can use `C-v` to insert and `C-c` to copy the selection.

2 XEmacs Support for ZeTa

ZETA can be operated from within a comfortable XEmacs mode, `zeta-source-mode`. To start a ZETA session from within XEmacs, issue `M-x zeta`.

The mode realizes a session log as described for the graphical user interface. The following snapshot represents the same operation state as in the previous section, where we just executed the expression *BirthdayBook*:

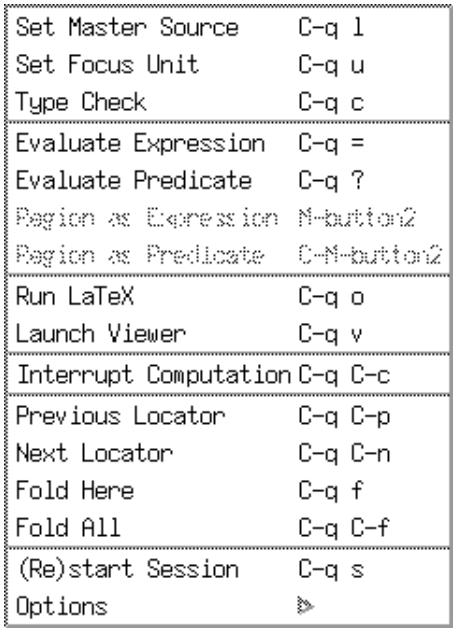


Clicking on the highlighted locaters visits the referred sources.

In the commander's buffer, you can type in ZETA commands. The set of available commands is displayed with the `help` command. `C-up` and `C-down` lets you step through the command history. In order to enter multiline arguments, quote them, as in

```
e "
   date \lbind day==1, month==2, year==1998 \rbind
 "
```

ZeTa's operations can be also reached via the `ZETA` menu and various key-shortcuts which all start with `C-q`:



The current default unit (section) to operate on is defined with `C-Q u`. If any of the commands which operate on sections, such as type checking, execution, or formatting are activtated via the menu or by a key-shortcut from within a source buffer, ZeTa tries to automatically determine the default unit by scanning for `\zsection` directives in the buffer.

A non-terminating operation can be interrupted using `C-q C-c`. This is in particular useful for execution, where non-termination is a possible behaviour.

3 Batch Operation

On startup, the ZeTa system reads and executs commands in the files `$ZETABASE/lib/zetarc`, `$HOME/.zetarc` and `./zetarc` in this order. The first file usually loads the adaptors. You can use any ZeTa command in the batch files, but note that some commands are not available unless the adaptors which provide them have been loader. Use the `help` command to get a list and short explanation of all available commands. Useful commands in zetarc-scripts are:

- `plugin <class>`: Plugin an adaptor specified by the given class. The class must be found in ZeTa's classpath.
- `load <filename>`*: load a LaTeXfile.
- `path <pathname>`*: set the LaTeXsearch path.
- `make <unitname> <content-type>`: establish the given content type for the unit.